SYMBIOSIS
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Transformer-Based Depression Detection from Social Media Text Using Contextual Language Representations

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Turning Social Media Insights into Mental Health Support

1. Introduction

Social-media timelines carry language and behavioural signals linked to depression. This study investigates how different transformer architectures and text representations affect user-level depression detection.

2. Research Questions

- **RQ1** How does text representation affect results?
- **RQ2** How does architecture affect results?
- **RQ3** Does the best representation depend on the architecture?
- **RQ4** How do sparse-attention architectures perform under a common 512-token input constraint?

3. Our Pipeline

Dataset (MDDL) 7,669 users → Preprocessing (4 Conditions) → 7 Transformer Models → Multimodal Fusion (Text + Metadata + Emotion) → Depression Prediction (D1 / D2)

Diagnostic-Anchor Removal: 1,920 Records (1,847 Pairs)

4. Experimental Setup

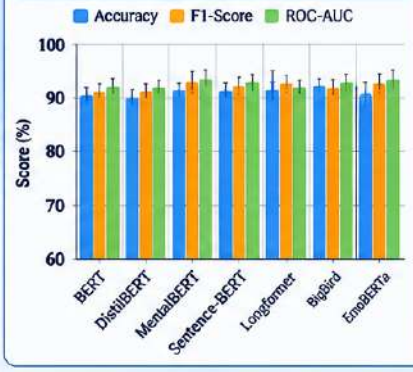
Dataset Details
MDDL (Matter about Depression)
Total Users : 7,669
Train : 6,135
Val : 767 | Test : 767

Four Representations
- Raw
- Aggressive
- Selective
- Emotion-Based

53 Metadata Features
7 Feature Groups:
Behavioural, Linguistic, Readability, Emotional, Psychological, Writing Style & LDA Topics

5. Key Findings

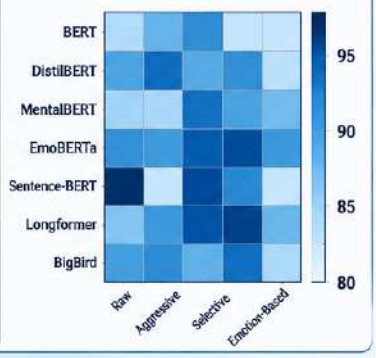
Performance Across Architectures



Top Results

- **Best F1-Score** 89.49% (Sentence-BERT + Raw)
- **Best Accuracy** 92.96% (Sentence-BERT + Raw)
- **Best ROC-AUC** 97.89% (Longformer + Selective)

F1-Score Heatmap



6. Model Insights

Explainability Methods
- SHAP – Feature Importance
- LIME – Local Explanation

t-SNE
- Representation Visualization

Grad-CAM-style Token Sensitivity
- Key Token Analysis

Key Takeaway
Emotion-Based and Selective representations improve performance.

7. Conclusion

Our study demonstrates that both the choice of transformer architecture and the text representation strategy significantly influence user-level depression detection. Sentence-BERT and Longformer show strong performance, highlighting the importance of model selection and preprocessing in mental health analytics.

Limitations & Responsible Use
Identifies computational patterns, not a clinical diagnostic tool. Results are based on one dataset and may not generalize to all populations.

Key Insight: Careful representation and model choice can significantly improve depression detection.

Towards responsible and empathetic AI for mental health